CS 4380 Project 4: Procedures and Beyond

Introduction

At this point in the semester, you have created a working emulator and assembler for the 4380, and have used it to investigate the implications of program behavior and caching. For this, the fourth and final project of the semester you will complete your emulator and assembler by implementing the 4380 Mark 4. The Mark 4 supports procedure calls and includes instructions to facilitate dynamic memory management (both features that you will find useful when utilizing your emulator and assembler in 4490). You will also add features that facilitate working with arrays and strings. With these added features in place you will exercise the full range of your assembler and emulator's capabilities by implementing several programs that will be described below. These programs begin simple and increase in complexity, with the final program being worth 50% of your final exam grade.

Part 1: 4380 Emulator and Assembler Mark 4

To meet the requirements of this lab it will be necessary to extend the 4380 emulator and assemblers with support for the following additional instructions:

Logical Instructions: (NOT BITWISE)

Value	Operator	Operand 1	Operand 2	Description
18	AND	RD	RS	Perform RD && RS, store result in RD
19	OR	RD	RS	Perform RD || RS, store result in RD

Arithmetic Instructions:

Value	Operator	Operand 1	Operand 2	Description
14	ADDI	RD	Imm*	Add Imm to RD, store the result in RD
33	MULI	RD	Imm*	Multiply RD by Imm, store the result in RD
34	DIVI	RD	Imm*	Divide RD by Imm (signed), result in RD
38	SDIV	RD	RS	Store result of signed division RD / RS in RD

* When represented in an assembly file the Imm value shall be represented as a numeric literal.

Procedure/Stack Instructions:

Value	Operator	Operand 1	Operand 2	Description
40	PSHR	RS	DC*	Place the word in RS onto the stack, set SP = SP - 4
41	PSHB	RS	DC*	Place the least significant byte in RS onto the stack; set SP = SP - 1
42	POPR	RD	DC*	Place the word on top of the stack into RD, update SP = SP + 4
43	POPB	RD	DC*	Place the byte on top of the stack into RD, update SP = SP + 1
44	CALL	Address	DC*	Push PC onto stack, update PC to Address
45	RET	DC	DC*	Pop stack into PC

*DC = "Don't Care". This operand shall be omitted in 4380 assembly instructions but is required in 4380 bytecode.

Heap Instructions: WAIT TO IMPLEMEMT...

Value	Operator	Operand 1	Operand 2	Description
35	ALCI	RD	Imm	
36	ALLC	RD	Address	
37	IALLC	RD	RS	

Taps/Interrupts:

Value	Operator	Operand 1	Operand 2	Description
21	TRP	#5	DC*	Write the Pascal-style string whose starting address is in R3 to stdout.
21	TRP	#6	DC*	Read a string from stdin and store it in memory as a Pascal-style string whose starting address is in R3.

*DC = "Don't Care". This operand shall be omitted in 4380 assembly instructions but is required in 4380 bytecode.

Directives:

Directive	Operand	Examples	Behavior
.BTS	Required unsigned decimal value. This value is the number of bytes to be allocated.	.BTS #25 .BTS #5 a_label .BTS #20	Allocates the specified number of bytes in place and initializes them all to 0. If an optional label is present the label shall be associated with the address of the first byte allocated.
.STR	Required double quote delimited string.	.STR "Example!" name .STR "Fred"	Allocates a number of bytes equal to the length of the string + 2 and initializes them to the ascii values for the characters in the string, preceded by the length of the string (0-255) and followed by the value 0 (null character). If an optional label is present the label shall be associated with the address of the first byte allocated.

Unless otherwise specified below - all requirements from projects 1 , 2, and 3 remain in force for the assembler and emulator. The following additional requirements must also be met.

Req1) Project 4 work must be completed in a separate branch called project-4. When all requirements are met, the branch should be merged into main (BUT NOT DELETED).

Req2) Upon startup, and after reading the input file into prog_mem, the 4380 emulator shall initialize SB to the highest available memory address. **Note:** SB can be changed by a program at runtime.

Req3) Upon startup, and after reading the input file into prog_mem, the 4380 emulator shall initialize SP to the highest available memory address. **Note:** SP can be changed by a program at runtime.

Req4) Upon startup, and after reading the input file into prog_mem, the 4380 emulator shall initialize SL to the address of the first byte past the last instruction read into program memory. **Note:** SL can be changed by a program at runtime.

Req5) The 4380 emulator shall not allow SP to be set to a value higher than SB, or lower than SL. If at any time a program attempts to set SP to a value outside the range of [SL, SB], the emulator shall print the message “INVALID INSTRUCTION AT: <address>” where <address> is the address in Program Memory of the first byte of the offending instruction. The emulator shall then exit with a return value of 1.

Part 2: Programming the 4380

Coming soon...